

Two Geometric Readings of a Thermodynamic Surface

Geometrothermodynamics and the Directive–Substrate–Product coupling
diagnostic, side by side, with an independently verified curvature

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Abstract

Both methods act on the *same* object: the Hessian of a thermodynamic fundamental relation $\Phi(E^a)$. **GTD** wraps that Hessian in a Legendre–covariant conformal factor and reads the *curvature scalar* of the resulting metric; its singularities mark the instability (Davies) locus. We show **DBP** admits a parallel construction — a metric assembled from the role–coupling channels, $g^{\text{DBP}} = -h^{-1}$ — whose curvature is likewise a single scalar, and that the two curvatures are *complementary*. On the Kerr black hole we verify in exact computer algebra that conventional GTD-II’s curvature has a double pole on the Davies transition and is provably blind to the extremal boundary; the DBP curvature does the opposite, diverging on *both* physical boundaries and finite on Davies. This complementarity is *universal*: across Kerr (spin), Reissner–Nordström (charge), and the van der Waals fluid (no reflection symmetry, a genuine critical point), GTD-II’s curvature carries an order–2 pole on the instability locus while the DBP curvature carries an order–3 pole on each role boundary — order 4 where a boundary meets a reflection–symmetry fixed locus — the two singular sets disjoint. The DBP geometry is geodesically incomplete, its boundaries finite–distance curvature edges. At $n = 3$ (Kerr–Newman) a two–state demonstration shows that both single scalars are lossy and the complete S_3 –invariant is a faithful three–dimensional carrier. Transcribed literature and independently verified computation are distinguished throughout by inline certainty tiers.

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1 Shared setup: one surface, one Hessian

A thermodynamic system with n degrees of freedom is fixed by a fundamental relation

$$\Phi = \Phi(E^1, \dots, E^n), \quad I_a = \frac{\partial \Phi}{\partial E^a}, \quad (1)$$

with Φ a potential, E^a the extensive variables and I_a their conjugates. The second-order data is the Hessian

$$H_{ab} = \frac{\partial^2 \Phi}{\partial E^a \partial E^b} = \text{Weinhold metric (energy representation)}. \quad (2)$$

Both methods below are functions of H_{ab} and the first derivatives I_a . They differ only in what they do with that data.

Running example (Kerr). The Kerr black hole has $n = 2$, extensive variables (S, J) (entropy, angular momentum), potential $\Phi = M$ (mass/energy), and Christodoulou relation

$$\boxed{M(S, J) = \sqrt{\frac{S}{4\pi} + \frac{\pi J^2}{S}}} \iff S = 2\pi \left(M^2 + \sqrt{M^4 - J^2} \right). \quad (3)$$

The first derivatives are the temperature and angular velocity, $M_S = T$, $M_J = \Omega$. Two physical boundaries bound the wedge $0 \leq J \leq S/2\pi$: the *Schwarzschild* edge $\Omega = 0$ ($J = 0$) and the *extremal* edge $T = 0$ ($S = 2\pi J$, i.e. $J = M^2$). Inside sits the *Davies* locus where the heat capacity $C_J = T/M_{SS}$ diverges, i.e. $M_{SS} = 0$, which evaluates to

$$48\pi^4 J^4 + 24\pi^2 J^2 S^2 - S^4 = 0 \iff \frac{J^2}{M^4} = 2\sqrt{3} - 3 \approx 0.4641. \quad (4)$$

Roles are charges. The role assignment used throughout is the physical one: the DBP *roles* are the conserved charges of the system, and the potential Φ is the graph value that the surface $F = 0$ expresses as a function of them. The faithfulness theorem of the role-channel companion is stated for exactly this assignment, and it is the same Hessian H_{ab} that GTD reads — which is what makes “same surface, two readings” a literal claim and not an analogy. A book-keeping rule

fixes the transition from $n = 2$ to $n = 3$: *energy is an honorary role only when the charges number fewer than three*. Kerr has two charges (S, J) , so mass fills the third role slot and the active S_3 acts on $\{S, J, M\}$; Kerr–Newman (§5) has three charges (S, J, Q) , so mass returns to its proper place as the graph value and S_3 acts on the three charges directly. The $n = 2$ worked example below should be read with this in mind — it is the degenerate case in which the potential is conscripted to make up the count.

2 Geometrothermodynamics (GTD): full pipeline

2.1 Contact structure and the equilibrium metric

GTD works on a $(2n+1)$ -dimensional phase space $\mathcal{T}$ with coordinates $Z^A = \{\Phi, E^a, I^a\}$, the Gibbs 1-form

$$\Theta = d\Phi - \delta_{ab} I^a dE^b, \quad (5)$$

and a metric G on $\mathcal{T}$ that is required to be invariant under Legendre transformations. The equilibrium manifold $\mathcal{E} \subset \mathcal{T}$ is the embedding $\varphi : \{E^a\} \mapsto \{\Phi(E^a), E^a, I^a(E^a)\}$ on which $\varphi^*\Theta = 0$ (the first law). The induced *equilibrium metric* is the pullback $g = \varphi^*(G)$.

2.2 The three conventional metrics

The metric forms, conformal factor, and the contact-geometry construction below are transcribed from the geometrothermodynamics literature. [literature] For the standard Legendre-invariant choices of G , the pullback gives (constants β_c ; sum over repeated indices)

$$\mathcal{G}^{\text{I}} = \left(\sum_c \beta_c E^c \partial_c \Phi \right) \frac{\partial^2 \Phi}{\partial E^a \partial E^b} dE^a dE^b, \quad (6)$$

$$\mathcal{G}^{\text{II}} = \left(\sum_c \beta_c E^c \partial_c \Phi \right) \eta_a^d \frac{\partial^2 \Phi}{\partial E^d \partial E^b} dE^a dE^b, \quad \eta = \text{diag}(-1, 1, \dots, 1), \quad (7)$$

$$\mathcal{G}^{\text{III}} = \left(\sum_a \beta_a E^a \partial_a \Phi \right) \frac{\partial^2 \Phi}{\partial E^a \partial E^b} dE^a dE^b. \quad (8)$$

Each is **(conformal factor) $\times$ (Hessian)**, optionally signature-twisted by η . The conformal factor $\omega = \sum_c E^c \partial_c \Phi = \sum_c E^c I_c$ (an Euler/Smarr combination) is what carries Legendre invariance.

2.3 The modified metrics (Mahanta, 2025–2026)

Conventional GTD reproduces second-order transitions but fails to encode the physical *boundaries*; this much is established in the cited literature, where the modified metrics are shown to recover the temperature-vanishing and spinodal curves. [literature] The proposed fix modifies (6)–(8) to

$$\mathcal{G}_{\text{mod}}^{\text{I}} = \left(\sum_c \beta_c E^c \partial_c \Phi \delta_1^c \right) \delta_a^b \frac{\partial^2 \Phi}{\partial E^a \partial E^b} dE^a dE^b, \quad (9)$$

$$\mathcal{G}_{\text{mod}}^{\text{III}} = \mathcal{G}^{\text{III}} \text{ but with off-diagonal elements deleted: } m_{sg} = 0. \quad (10)$$

Read mechanically, the modification is two surgeries: **(i)** the δ_1^c shrinks the conformal factor from the full Euler sum to its first term, so it *vanishes on a boundary*; **(ii)** the off-diagonal coupling is *amputated*, $m_{sg} = 0$. The second reading is the rhetorical hinge of this paper—it is the quantity DBP measures (§3)—and is stated here as our reading of the modified construction; the explicit $m_{sg} = 0$ form should be confirmed against the metric equations of Mahanta (2025). [literature: confirm vs. source] Nothing in the side-by-side comparison or in the $n = 3$ results depends on this

reading: the modified metrics enter only as the literature’s own account of why conventional GTD needs patching to see boundaries, which is the foil against which the DBP boundary–detection (§3.4) is contrasted. The load–bearing GTD computation in this paper is the *conventional* GTD-II curvature of §2.4 and §5, not any modified metric.

2.4 Worked Kerr: metric $\rightarrow$ curvature $\rightarrow$ singularity (verified)

Take $\mathcal{G}^{\text{II}}$ with $\eta = \text{diag}(-1, 1)$. The conformal factor obeys the Smarr relation $M = 2(TS + \Omega J)$, so

$$\omega = SM_S + JM_J = TS + \Omega J = \frac{1}{2}M. \quad (11)$$

The (diagonal) equilibrium metric is

$$g^{\text{II}} = \omega \text{diag}(-M_{SS}, M_{JJ}), \quad g_{SS} = \frac{S^4 - 24\pi^2 J^2 S^2 - 48\pi^4 J^4}{32\pi S^3(4\pi^2 J^2 + S^2)}, \quad g_{JJ} = \frac{\pi S}{2(4\pi^2 J^2 + S^2)}, \quad (12)$$

and the displayed components agree with $g_{SS} = -\omega M_{SS}$, $g_{JJ} = \omega M_{JJ}$ ($\omega = M/2$) by direct computation. [computer-verified] Note already that the numerator of g_{SS} is $-(48\pi^4 J^4 + 24\pi^2 J^2 S^2 - S^4)$: the metric component itself vanishes on the Davies locus.

For a diagonal surface metric $ds^2 = E dS^2 + G dJ^2$ the Ricci scalar is $R = 2K$ with Gaussian curvature

$$K = -\frac{1}{2\sqrt{EG}} \left[\partial_S \left(\frac{\partial_S G}{\sqrt{EG}} \right) + \partial_J \left(\frac{\partial_J E}{\sqrt{EG}} \right) \right]. \quad (13)$$

We evaluated (13) for g^{II} in exact computer algebra. The result is a rational function of (S, J) whose denominator factors exactly as

$$\boxed{\text{den } K = S (48\pi^4 J^4 + 24\pi^2 J^2 S^2 - S^4)^2} \quad (14)$$

a *double* pole on the Davies locus (4), $M_{SS} = 0$, and **nothing else**. [computer-verified] Two facts follow, both checked exactly rather than asserted:

- (i) **K diverges precisely on the Davies locus**—the GTD “success”: the curvature singularity coincides with the second-order phase transition.
- (ii) the extremal factor $(S^2 - 4\pi^2 J^2)$, i.e. $T = 0$, **does not divide** (14): conventional GTD-II is *blind to the extremal boundary*. [computer-verified]

This is exactly the deficiency the modified metrics of §2.3 were built to patch, and it is the place where the DBP construction differs sharply: the same extremal boundary appears for free in DBP as the role-singular locus $a = M_S = T = 0$ (§3.4), with no metric and no conformal engineering.

GTD pipeline, in one line

$$\Phi(E^a) \xrightarrow{\partial^2, \omega} g_{ab} = \omega \eta_a^c H_{cb} \xrightarrow{\Gamma, \text{Riem}} R[g] \xrightarrow{\text{singularities}} \text{phase transitions.}$$

3 Directive–Substrate–Product (DBP): full pipeline

All DBP equations in this section—the S_3 relations, the coupling channels, and the three forms of $\mathcal{A}_c$ —are the foundational role-jet constructions of [1], established there by exact computer algebra; the Kerr results below appear in the companion “Role-Channel Anisotropy of DBP Surfaces.” Both are reproduced here only to set the comparison. [computer-verified]

3.1 Surface, roles, and the active S_3

DBP treats the fundamental relation as a constraint surface $F(D, S, P) = 0$ with three *roles*: Directive, Substrate, Product. Locally one solves for an output as a graph, e.g. $P = f(D, S)$, with finite 2-jet

$$(a, b, A, B, C) = (f_D, f_S, f_{DD}, f_{DS}, f_{SS}). \quad (15)$$

Role permutation is realised *actively*: re-solving $F = 0$ for a different output and transporting the jet through the inverse maps. To first order the generators are

$$s(a, b) = (b, a) \quad (\text{input swap}), \quad t(a, b) = \left(\frac{1}{a}, -\frac{b}{a}\right) \quad (\text{output transposition } P \leftrightarrow D), \quad (16)$$

and they satisfy

$$\boxed{s^2 = t^2 = (st)^3 = e} \implies \langle s, t \rangle \cong S_3, \quad (17)$$

a genuine birational S_3 action on the regular jet locus ($ab \neq 0$). *This group is the structural object GTD lacks* — discrete Legendre transformations do not form a group.

3.2 Output–role coupling channels

For each of the three role–outputs, the off–diagonal (coupling) coefficient of the second fundamental form, transported into that chart, is

$$\Lambda_P = B, \quad \Lambda_D = \frac{Ab - aB}{a}, \quad \Lambda_S = \frac{Ca - bB}{b}. \quad (18)$$

Normalising by $q_0 = 1 + a^2 + b^2$ gives the three coupling curvatures

$$\kappa_{c,P} = -\frac{\Lambda_P^2}{q_0^2}, \quad \kappa_{c,D} = -\frac{\Lambda_D^2}{q_0^2}, \quad \kappa_{c,S} = -\frac{\Lambda_S^2}{q_0^2}. \quad (19)$$

The unordered triple $\{\kappa_{c,P}, \kappa_{c,D}, \kappa_{c,S}\}$ is an S_3 –orbit: role permutation permutes the three values.

3.3 The coupling anisotropy $\mathcal{A}_c$

The diagnostic is the simplest non–constant symmetric (hence S_3 –invariant) function that detects collapse of that orbit — the sum of squared pairwise gaps:

$$\boxed{\mathcal{A}_c = (\kappa_{c,P} - \kappa_{c,D})^2 + (\kappa_{c,D} - \kappa_{c,S})^2 + (\kappa_{c,S} - \kappa_{c,P})^2} \quad (20)$$

Three equivalent readings:

1. **Statistical:** $\mathcal{A}_c = 9 \text{Var}(\kappa_{c,P}, \kappa_{c,D}, \kappa_{c,S})$ — the variance of the coupling curvature over the role orbit (the classical identity that the summed squared pairwise differences are proportional to the variance; the absolute-difference analogue is Gini’s mean difference, 1912).
2. **Representation–theoretic:** the channel vector lives in the permutation representation $\mathbb{R}^3 = \mathbf{1} \oplus \mathbf{2}$ of S_3 (trivial $\oplus$ standard). Writing P_2 for projection onto the 2–dimensional standard irrep,

$$\mathcal{A}_c = 3 \|P_2 \kappa_c\|^2. \quad (21)$$

The trivial part $\mathbf{1}$ is the role–blind *mean* coupling; $\mathcal{A}_c$ is the squared norm of everything that distinguishes the roles. Isotropy $\mathcal{A}_c = 0$ is “purely $\mathbf{1}$ ”, a codimension–2 condition because $\dim \mathbf{2} = 2$.

3. Explicit (jet) form:

$$\begin{aligned}
\mathcal{A}_c &= \frac{D_{PD}^2 + D_{DS}^2 + D_{SP}^2}{q_0^4}, & D_{PD} &= \frac{Ab(2aB - Ab)}{a^2}, \\
D_{DS} &= \frac{(Ab^2 - Ca^2)(Ab^2 + Ca^2 - 2abB)}{a^2b^2}, & D_{PD} + D_{DS} + D_{SP} &\equiv 0. \\
D_{SP} &= \frac{Ca(Ca - 2bB)}{b^2},
\end{aligned} \tag{22}$$

The telescoping identity is the statement that the gap vector has no **1** component.

3.4 Worked Kerr: $\mathcal{A}_c$, the Davies reconciliation, and the boundaries

For Kerr, the graph is $M = f(S, J)$ (3) with role assignment $(D, S, P) = (\text{entropy } S, \text{momentum } J, \text{mass } M)$; the jet is $(a, b, A, B, C) = (M_S, M_J, M_{SS}, M_{SJ}, M_{JJ})$.¹ The output charts are physically named:

P -output = mass = energy rep (Weinhold); D -output = entropy rep (Ruppeiner); S -output = J -rep.

Two exact results, both verified symbolically:

Result 1 (Davies = Weinhold/Ruppeiner reconciliation). At the Davies locus $A = M_{SS} = 0$ (where C_J diverges, $j^2 = 2\sqrt{3} - 3$), the generic identity $\Lambda_D = -\Lambda_P$ holds, so

$$\frac{\Lambda_D}{\Lambda_P} = -1, \quad \kappa_{c,D} = \kappa_{c,P}, \quad \frac{\Lambda_S}{\Lambda_P} = -(1 + \sqrt{3}).$$

The energy (Weinhold) and entropy (Ruppeiner) coupling curvatures become *exactly equal in magnitude* at the phase transition; the third (J -) representation stays distinct. This is *partial* isotropy: the coupling vector crosses the $\kappa_{c,P} = \kappa_{c,D}$ reflection axis of the standard irrep. Full isotropy $\mathcal{A}_c = 0$ is empty in the physical wedge.

Result 2 (Boundaries are role-chart singularities, for free). The role-singular loci of the construction are

$$a = M_S = T = 0 \Rightarrow S = 2\pi J \text{ (extremal)}, \quad b = M_J = \Omega = 0 \Rightarrow J = 0 \text{ (Schwarzschild)}.$$

These are the two physical boundaries of the Kerr wedge. They appear because $\Lambda_D \propto 1/a$ and $\Lambda_S \propto 1/b$ *diverge* when an output derivative vanishes — one literally cannot solve for that variable as output. No engineering of a conformal factor is required: the boundaries are forced by the S_3 structure.

DBP pipeline, in one line

$$F(D, S, P) = 0 \xrightarrow{\text{jet } (a, b, A, B, C)} \{\Lambda_P, \Lambda_D, \Lambda_S\} \xrightarrow{\kappa_{c,i} = -\Lambda_i^2/q_0^2} \mathcal{A}_c = 3\|P_2\kappa_c\|^2.$$

¹The subscript S on $\Lambda_S, \kappa_{c,S}$ denotes the *substrate role* (the momentum-output chart), not the entropy variable S .

4 Side-by-side

	GTD	DBP
Raw object	Hessian $H_{ab} = \partial_a \partial_b \Phi$	same Hessian H_{ab}
Structure group	none (Legendre maps don't close)	S_3 (genuine: $s^2=t^2=(st)^3=e$)
Primary construction	metric $g_{ab} = \omega \eta_a^c H_{cb}$	coupling channels $\Lambda_P, \Lambda_D, \Lambda_S$
Normalisation	conformal factor $\omega = \sum_c E^c I_c$	$q_0 = 1 + a^2 + b^2$
Diagnostic	curvature scalar $R[g]$	coupling variance $\mathcal{A}_c = 3\ P_2 \kappa_c\ ^2$
Invariance mechanism	engineer ω, η so R is Legendre-invariant	$\mathcal{A}_c$ symmetric in the orbit $\Rightarrow$ S_3 -invariant by construction
Treatment of coupling	curvature <i>of</i> it; “fix” <i>deletes</i> m_{sg}	<i>measures its frame-variance</i> (the deleted quantity)
Transition signal (Kerr)	$R \rightarrow \infty$ at Davies $M_{SS}=0$	$\kappa_{c,P}=\kappa_{c,D}$ at Davies (W=R)
Boundary detection	must be engineered (modified ω)	automatic: role-chart singularities $a=0, b=0$
What it outputs	a number R at each state	a number $\mathcal{A}_c$ at each state
Has metric tensor?	yes	yes: channel-inverse $g^{\text{DBP}} = -h^{-1}$ (§6.1)
Has connection / geodesics?	yes	yes: geodesically incomplete; boundaries at finite distance (§6.4)
Has curvature scalar?	yes (R ; sees Davies)	yes ($R[g^{\text{DBP}}]$; sees the boundaries — <i>complementary</i> , §6.3)
Has invariant ring?	partial	resolved: faithful 3-dim carrier O ; $\mathcal{A}_c$ its lowest coordinate (§5)

5 The $n = 3$ case: Kerr–Newman

The $n = 2$ comparison above has a structural soft spot: with only two charges, the potential is conscripted as a role, and the three role-outputs are not on equal physical footing. Kerr–Newman removes the crutch. It has three charges (S, J, Q) — entropy, angular momentum, electric charge — so by the book-keeping rule of §2.2 the mass returns to being the graph value and the active S_3

permutes three genuine charges. The fundamental relation is the Smarr/Christodoulou form

$$\boxed{M(S, J, Q) = \sqrt{\frac{S}{4\pi} + \frac{\pi J^2}{S} + \frac{Q^2}{2} + \frac{\pi Q^4}{4S}}} \xrightarrow{Q \rightarrow 0} \text{Kerr (3)}, \quad (23)$$

[literature] with the polynomial (rationalised) form

$$F = S^2 + 2\pi S (Q^2 - 2M^2) + \pi^2 (4J^2 + Q^4) = 0. \quad (24)$$

The two extremal/boundary conditions generalise: the temperature vanishes ($T = M_S = 0$) on

$$S = \pi \sqrt{4J^2 + Q^4} \iff M^4 = J^2 + M^2 Q^2, \quad (25)$$

which reduces to the Kerr extremal edge $S = 2\pi J$, $J = M^2$ at $Q = 0$. [computer-verified]

5.1 GTD-II for Kerr–Newman: again one pole, again blind

Take $\mathcal{G}^{\text{II}}$ with $\eta = \text{diag}(-1, 1, 1)$ and conformal factor $\omega = SM_S + JM_J + QM_Q$. The signature twist sets the mixed weight $\frac{1}{2}(\eta_{aa} + \eta_{cc})$ to zero on every pair involving the first (S) slot, so the metric is *block* structured:

$$g^{\text{II}} = \omega \begin{pmatrix} -M_{SS} & 0 & 0 \\ 0 & M_{JJ} & M_{JQ} \\ 0 & M_{JQ} & M_{QQ} \end{pmatrix}, \quad (26)$$

the entropy direction decoupling while (J, Q) remain coupled. The (J, Q) Hessian block is nondegenerate on the physical domain — its determinant is proportional to $(\pi Q^2 + S)^3 > 0$ for $S > 0$ — so the only place the curvature can blow up is the first slot. [computer-verified]

We evaluated the 3-dimensional Ricci scalar $R[g^{\text{II}}]$ in numerically-exact computer algebra (the Ricci routine validated on flat space $R=0$ and the unit 2-sphere $R=2$). The outcome is the exact $n = 3$ analogue of the Kerr result: R has a *single* physical curvature singularity, a *double* pole on the $n = 3$ Davies locus $M_{SS} = 0$, whose vanishing factor is

$$\boxed{\mathcal{D}_3 = 48\pi^4 J^4 + 24\pi^4 J^2 Q^4 + 32\pi^3 J^2 Q^2 S + 24\pi^2 J^2 S^2 + 3\pi^4 Q^8 + 8\pi^3 Q^6 S + 6\pi^2 Q^4 S^2 - S^4} \quad (27)$$

which reduces to the Kerr Davies quartic (4) at $Q = 0$. [computer-verified] Two facts carry over verbatim, both checked rather than asserted:

- (i) $R \sim 1/\varepsilon^2$ as $\mathcal{D}_3 \rightarrow 0$: the curvature singularity again coincides with the heat-capacity (Davies) transition, the GTD “success.”
- (ii) R is *finite* on the extremal locus (25) (numerically $R \approx -33.6$ at a representative extremal point): the extremal factor does not divide $\text{den } R$, so conventional GTD-II is *blind to extremality* at $n = 3$ exactly as at $n = 2$. [computer-verified]

The full factored closed form of $\text{den } R$ — the $n = 3$ analogue of $\text{den } K = S(\text{Davies})^2$ in (14) — is

$$\boxed{\text{den } R = [\text{exact closed form; } \mathcal{D}_3^2 \text{ is its only physical vanishing factor — value pending verification run]}} \quad (28)$$

and will be slotted in once the exact factorisation completes; the structural facts (i)–(ii) above do not depend on it.

5.2 DBP for Kerr–Newman: the faithful carrier

With (S, J, Q) the three roles and M the graph value, each of the three charge–output charts carries its own coupling channel exactly as in §3, and the unordered triple is again an S_3 –orbit. Evaluated at a regular interior point $(S, J, Q) = (5, \frac{1}{2}, \frac{2}{5})$ the three chart denominators are the three physical response functions

$$S\text{-output: } g_1 = T \text{ (Directive),} \quad J\text{-output: } g_2 = \Omega \text{ (Substrate),} \quad Q\text{-output: } g_3 = \Phi_e \text{ (Product),}$$

and the extremal locus (25) is, once again *for free*, the role–chart singularity where the entropy–output channels diverge as $g_1 = T \rightarrow 0$. The same boundary GTD-II’s R cannot see is a coordinate singularity of the DBP construction. [symbolic]

The new content at $n = 3$ is that the single scalar is no longer enough — and DBP already carries the object that is. The complete S_3 –invariant of the coupling is not $\mathcal{A}_c$ but the faithful *three–dimensional* role–channel carrier $O = (O_{12}, O_{13}, O_{23})$ (the off–diagonal RoleChSpec coordinates); at the point above, $O = (-0.162, -0.086, -0.856)$. [computer-verified]

Result 3 (The invariant content is a faithful 3–dimensional carrier; $\mathcal{A}_c$ is its lowest piece). The role–channel map sends a coupling state to the orbit carrier $O \in \mathbb{R}^3$, and this map is *faithful*: the 6×3 rank–one channel matrix has rank 3, with an explicit minor certificate $32/(g_1 g_2 g_3) = 2^5/(g_1 g_2 g_3)$ in characteristic $\neq 2$ and a separate characteristic–2 certificate, so the carrier is 3–dimensional in *every* characteristic; the faithfulness of the coupling carrier is established in the foundational calculus [1]. The scalar $\mathcal{A}_c$ is the squared norm of the standard–irrep part of O — the *lowest* non–trivial S_3 –invariant. It is therefore a genuine but lossy summary: distinct carriers can share an $\mathcal{A}_c$ value. This resolves what an earlier draft posed as an open “invariant ring” target — the complete invariant content is the carrier itself, and $\mathcal{A}_c$ is one coordinate of it.

5.3 Two states, one R , one $\mathcal{A}_c$, two carriers

The promotion above has teeth only if the loss is real. It is, and the demonstration is the centrepiece of the $n = 3$ case. By pseudo–arclength continuation along the simultaneous level set $\{R = \text{const}\} \cap \{\mathcal{A}_c = \text{const}\}$ we located two *physically distinct* regular Kerr–Newman states

$$p_1 = (S, J, Q) = (12, 0.6, 0.5), \quad p_2 = (13.7965, 0.6954, 0.5589),$$

on which the two scalar diagnostics are *each* constant to machine precision:

$$R(p_1) = R(p_2) = 269.3252, \quad \mathcal{A}_c(p_1) = \mathcal{A}_c(p_2) = 297.1150, \quad |\Delta| \approx 10^{-13}, \quad (29)$$

yet whose carriers differ, $\|O(p_1) - O(p_2)\| \approx 0.047$, and whose six measured rank–one channels separate by 9%. [computer-verified] Both single–scalar diagnostics — GTD’s curvature R and DBP’s own $\mathcal{A}_c$ — collapse a pair that the faithful three–dimensional carrier cleanly distinguishes.

Remark 1 (Where the honesty is). The contrast here is sharp but narrow, and it is worth stating exactly where it bites. It is a statement about the two pipelines *at the scalar diagnostic level* — the level at which each pipeline, as practised, actually terminates and reports a number. It is *not* the claim that GTD’s full metric g^{II} is degenerate on this pair (it is a nine–component object and we have not shown it coincides), and it is pointedly not a claim that DBP is immune: $\mathcal{A}_c$ fails on exactly the same pair. $n = 3$ is simply the first place where the single–scalar summary — whichever pipeline produces it — is provably incomplete, and where DBP happens to retain the faithful object one rung below its own headline scalar.

The loop closes on the boundary GTD misses. The S -output chart’s singularity at $g_1 = T = 0$ is not an isolated coincidence: it is the ramification of the entropy re-solving cover of the surface (24), and that ramification locus *is* the extremal boundary (25). In the language of the monodromy companion, the role boundaries are the branch locus of the re-solving cover and the local DBP invariant is singular exactly there; the entropy cover ramifies precisely on extremality, which is the one physical boundary the conventional GTD-II curvature R of §5.1 does *not* register. The $1/g_y$ normalisation of each role channel is the local shadow of that cover’s ramification divisor. This is the single forward pointer to the unification thread; it is developed there and not pursued further here.

6 The DBP metric, its curvature, and the complementarity law

Earlier drafts posed, as open “derivation targets,” the objects DBP lacked relative to GTD: a metric, a connection with geodesics, and a *single* intrinsic curvature scalar to set beside GTD’s R . For $n = 2$ all three now exist in closed form. The curvature scalar, moreover, does something conventional GTD-II’s cannot — it is singular exactly on the physical boundaries — and the relationship between the two curvatures turns out to be a universal complementarity. The construction is driven by one fact established in the role-channel companion: written in ambient form, the coupling coefficient is a directional curvature,

$$N_{ab} = (F_b, -F_a) \text{Hess}_{ab}(F) (F_b, -F_a)^\top = F_a^2 F_{bb} - 2F_a F_b F_{ab} + F_b^2 F_{aa}, \quad (30)$$

a chart-independent object depending only on the variable pair $\{a, b\}$; the output variable enters only through a normalisation $1/F_y$. [symbolic] Because N_{ab} carries no chart choice, the active S_3 acts on it by plain permutation of ambient labels — which makes an S_3 -covariant metric straightforward to write down, where the birational action on the jet had made it look hard.

6.1 The channel-inverse metric

The canonical S_3 -symmetric symmetric 2-tensor built from the channels pairs each role’s coupling curvature $\kappa_{c,i} = -\Lambda_i^2/q_0^2$ with its own coordinate differential and restricts to the surface tangent plane:

$$h = \sum_{i \in \{D, S, P\}} \kappa_{c,i} dE^i \otimes dE^i. \quad (31)$$

Every $\kappa_{c,i} < 0$, so h is negative-definite. The role boundaries are where an output becomes unsolvable and the corresponding channel *diverges* ($\Lambda_D \propto 1/a$, $\Lambda_S \propto 1/b$), so h blows up there; the object that *degenerates* on the boundaries is therefore its inverse. The DBP metric is

$$\boxed{g^{\text{DBP}} = -h^{-1}, \quad h = \sum_i \kappa_{c,i} dE^i \otimes dE^i,} \quad (32)$$

the sign flip making g^{DBP} positive-definite. The channel-to-coordinate pairing privileges no role, so g^{DBP} is S_3 -symmetric by construction — settling condition (i) of the old Target 1 at the level of the definition, not by a later check.

Result 4 (The DBP metric exists, is Riemannian, and is singular on both boundaries). On Kerr, $g^{\text{DBP}} = -h^{-1}$ is positive-definite throughout the physical wedge, with no interior degeneracy at any point checked. [computer-verified] Each role’s diagonal component vanishes on the boundary of its own role: $g_{SS}^{\text{DBP}} \rightarrow 0$ as $T = M_S \rightarrow 0$ (extremal) and $g_{JJ}^{\text{DBP}} \rightarrow 0$ as $\Omega = M_J \rightarrow 0$ (Schwarzschild).

On the extremal edge the determinant degenerates, $\det g^{\text{DBP}} \propto (S - 2\pi J)^2$ [computer-verified]; on the Schwarzschild edge the conjugate component diverges instead, so the metric is singular there as a component pole while the determinant stays finite. The asymmetry is structural: Schwarzschild ($J = 0$) is the fixed locus of the reflection $J \rightarrow -J$, which forces the cross-derivative $M_{SJ} \rightarrow 0$ and converts the degeneration into a pole. Both boundaries are singular loci of g^{DBP} .

6.2 The curvature scalar, and where it diverges

The Gaussian curvature $R[g^{\text{DBP}}]$ was computed in numerically-exact arithmetic (Brioschi form; the routine validated on flat space, $R=0$, and the unit sphere, $R=2$). A metric's component singularities can be coordinate artifacts — but Gaussian curvature is a coordinate invariant, so a divergence in it is geometrically real, which makes this the decisive test.

Result 5 (The DBP curvature detects the boundaries GTD-II misses — with exact leading coefficients). $R[g^{\text{DBP}}]$ *diverges* on both physical boundaries and is *finite* on the Davies transition — the exact mirror of conventional GTD-II. Every coupling channel Λ_i carries the single common factor $\sqrt{M^2}$ with $M^2 = S/4\pi + \pi J^2/S$ rational, so g^{DBP} — and hence $R[g^{\text{DBP}}]$ — is a *rational* function of (S, J, π) ; a one-variable Laurent expansion at each edge then gives the leading divergence in closed form:

$$\begin{aligned} R[g^{\text{DBP}}] &\sim \frac{C_{\text{ext}}(J)}{(S - 2\pi J)^3}, & C_{\text{ext}}(J) &= \frac{2(8J + 1)}{\pi(4J + 1)^3} > 0 \quad (\text{extremal}); \\ R[g^{\text{DBP}}] &\sim \frac{C_{\text{sch}}(S)}{J^4}, & C_{\text{sch}}(S) &= -\frac{96\pi S}{(16\pi S + 1)^2} < 0 \quad (\text{Schwarzschild}), \end{aligned}$$

with $R[g^{\text{DBP}}] \rightarrow -0.043$ finite on Davies [computer-verified]. The two closed-form coefficients and the pole orders 3 (extremal) and 4 (Schwarzschild) are [symbolic]: each was obtained three independent ways that agree — a hand telescoping of the Brioschi numerator at the edge, the full chain-rule determinant, and a fresh high-precision numeric Brioschi whose ratio to the closed form $\rightarrow 1$ as the edge is approached (checked at $J \in \{1, 2, 3\}$ for the extremal coefficient, $S \in \{5, 10, 20\}$ for the Schwarzschild coefficient). The extremal order reconciles structurally: $\det g^{\text{DBP}} \propto (S - 2\pi J)^2$, so $(\det g^{\text{DBP}})^2$ is order 4 and the Brioschi numerator supplies one compensating factor, $4 - 1 = 3$.

The two coefficients telescope to different depths, and the gap is itself a reflection signature. Each channel collapses at the boundary native to its role: the substrate channel $g_{JJ}^{\text{DBP}} \rightarrow 0$ at Schwarzschild ($\Omega = M_J = 0$), the directive channel $g_{SS}^{\text{DBP}} \rightarrow 0$ at extremal ($T = M_S = 0$). At Schwarzschild the curvature coefficient collapses to $C_{\text{sch}} = -6/g_2$, where $g_2 = \lim_{J \rightarrow 0} g_{JJ}^{\text{DBP}}/J^2$: it is fixed *entirely* by the collapse rate of the native channel, while the transverse channel g_{SS}^{DBP} — which carries a pole there, $g_{SS}^{\text{DBP}} \sim J^{-2}$ — cancels out of the curvature completely. At the extremal boundary no such cancellation occurs: $C_{\text{ext}} = g_1/(e_2 g_0)$ retains both the native collapse rate e_2 (from $g_{SS}^{\text{DBP}} \sim e_2(S - 2\pi J)^2$) and the transverse channel's value and slope g_0, g_1 (from $g_{JJ}^{\text{DBP}} = g_0 + g_1(S - 2\pi J) + \dots$). So the reflection symmetry does more than lift the pole order $3 \rightarrow 4$: at its fixed locus it *decouples the transverse channel from the leading coefficient*, leaving the curvature divergence governed by the native channel alone. [symbolic]

This closes the old Target 3 on GTD's own terms. Where conventional GTD-II's curvature is provably blind to the extremal boundary (§2.4), the DBP curvature diverges there; both are single intrinsic scalars built from the same Hessian.

6.3 The complementarity is universal

The Kerr result is one instance of a structural law. We ran the identical pair of constructions on two further surfaces, with the same fundamental–relation discipline (roles are charges; energy is the graph value): *Reissner–Nordström*, $M(S, Q) = (S + \pi Q^2)/2\sqrt{\pi S}$ (charge in place of spin), and the *van der Waals fluid*, $U(S, V) = e^{S/c}(V - b)^{-1/c} - a/V$ (internal energy, roles (S, V, U)) — which has *no* reflection symmetry and a genuine critical point. In every case the two curvatures’ singular sets are disjoint, and the orders transfer.

	Kerr (spin)	RN (charge)	vdW (fluid)
GTD-II curvature diverges on ...	Davies	Davies	spinodal $U_{VV}=0$
pole order	2	2	2
DBP curvature diverges on ... (generic boundary)	extremal	extremal	$P=0$
pole order	3	3	3
DBP curvature diverges on ... (reflection–fixed boundary)	$J=0$	$Q=0$	—
pole order	4	4	—
singular sets disjoint / complementary?	yes	yes	yes

Result 6 (Complementarity and the order law). For all three surfaces the conventional GTD-II curvature carries a *double pole* (order 2) on the instability/transition locus — Davies for the black holes, the adiabatic spinodal $U_{VV} = 0$ for van der Waals — and is finite on the role boundaries. The DBP channel–inverse curvature is finite on the transition and carries an *order–3* pole on each role boundary, *promoted to order 4* wherever a boundary coincides with a reflection–symmetry fixed locus (the non–rotating / uncharged edge $J = 0$, $Q = 0$, where $M_{SJ} \rightarrow 0$). The two singular sets are disjoint. [computer-verified] For Kerr both pole orders and their leading coefficients are now in closed form (Result 5), [symbolic]; the transfer of the order law to Reissner–Nordström and van der Waals remains [computer-verified].

The order–4 poles at $J = 0$ and $Q = 0$ are thus not the default behaviour but the *signature of the reflection symmetry*: van der Waals, lacking it, gives the generic order 3. The complementarity itself — each curvature blind exactly where the other is singular — is independent of that symmetry and holds on gravitational and classical surfaces alike. This is the sharp form of the paper’s thesis: the same Hessian supports two metrics whose curvature singularities *partition* the diagnostic loci between the transition and the boundaries.

6.4 Geodesics: the boundaries are finite–distance edges

With g^{DBP} in hand the connection and geodesics of the old Target 2a are immediate. The geodesic flow integrates cleanly — the affine speed $g_{ij}^{\text{DBP}} \dot{x}^i \dot{x}^j$ is conserved to $\sim 10^{-11}$ along the integrated curves, which validates the Christoffel computation. [computer-verified] The proper distance from

a representative interior state to either boundary is *finite* (extremal ≈ 25.2 , Schwarzschild ≈ 10.7): the DBP thermodynamic geometry is geodesically *incomplete*, its boundaries genuine finite-distance curvature edges rather than points at infinity — consistent with the curvature blowing up on approach.

6.5 What remains open

Two directions are not closed by the above.

Open — the holonomy route (old Target 2b).

The maps s, t are transports between the three role charts; treated as a discrete connection, their holonomy around a loop *encircling a boundary* is the monodromy of the re-solving cover around its ramification locus (the role-boundary), which is the wreath machinery of the monodromy companion. The earlier conjecture “holonomy = $\mathcal{A}_c$ ” cannot hold literally — $\mathcal{A}_c$ is a continuous scalar, the monodromy a discrete group element — and the honest statement is a bridge relating $\mathcal{A}_c$ ’s residue near a boundary to the cycle type of the monodromy there. This is the same thread as the companion and is not pursued further here.

Open — the $n = 3$ lift.

The metric $g^{\text{DBP}} = -h^{-1}$ and the complementarity above are established for $n = 2$. At $n = 3$ (Kerr–Newman, §5) each output chart carries *several* off-diagonal couplings rather than one, so the channel form h must be generalised before the construction transfers. Whether the complementarity survives against the $n = 3$ Davies surface is the natural next test.

The one-sentence statement. GTD and DBP read the same Hessian: GTD builds a metric whose curvature is singular on the instability locus and blind to the boundaries; DBP builds a metric whose curvature is finite on the instability locus and singular on the boundaries — and across spin, charge, and fluid these two singular sets are disjoint, with a universal order law (2 for the transition, 3 for a boundary, 4 for a reflection-fixed boundary).

A Verification protocol and provenance of each claim

This document mixes transcribed literature with independently verified computation; the tiers above mark which is which, and the table below summarises provenance.

Claim	Status
GTD contact structure, equilibrium metric, conventional $\mathcal{G}^{\text{I,II,III}}$	[literature] (Quevedo 2007, 2010)
Modified GTD metrics recover temperature-vanishing / spinodal boundaries	[literature] (Mahanta 2025, 2026)
Modified metric deletes the off-diagonal coupling ($m_{sg} = 0$)	[literature: confirm vs. source] (our reading; verify against Mahanta 2025 equations)
GTD-II Kerr metric components $g_{SS} = -\omega M_{SS}, g_{JJ} = \omega M_{JJ}$	[computer-verified] (exact CAS)
GTD-II Kerr curvature denominator $= S(48\pi^4 J^4 + 24\pi^2 J^2 S^2 - S^4)^2$	[computer-verified] (exact CAS; double pole on Davies)
GTD-II blind to the extremal boundary (extremal factor absent from den K)	[computer-verified] (exact CAS)
DBP S_3 relations $s^2 = t^2 = (st)^3 = e$	[symbolic]
DBP coupling channels, $\mathcal{A}_c$ three equivalent forms, telescoping	[symbolic] / [computer-verified]
Kerr Davies point $J^2/M^4 = 2\sqrt{3} - 3$; quartic $48\pi^4 J^4 + 24\pi^2 J^2 S^2 - S^4 = 0$	[computer-verified] (exact radicals)
Davies reconciliation $\Lambda_D/\Lambda_P = -1$, $\Lambda_S/\Lambda_P = -(1 + \sqrt{3})$	[symbolic]
Role-singular boundaries: extremal $S = 2\pi J$ ($a = 0$), Schwarzschild $J = 0$ ($b = 0$)	[symbolic]
$n = 3$ (Kerr–Newman)	
KN fundamental relation (23)–(24); $Q \rightarrow 0$ recovers Kerr	[literature] / [computer-verified] (exact CAS)
KN extremal locus $S = \pi\sqrt{4J^2 + Q^4}$, $M^4 = J^2 + M^2 Q^2$	[computer-verified] (exact CAS)
GTD-II KN metric block structure (S decoupled; (J, Q) block $\det \propto (\pi Q^2 + S)^3$)	[computer-verified] (exact CAS)
GTD-II KN curvature: double pole on $n = 3$ Davies $\mathcal{D}_3$ (27); reduces to Kerr at $Q=0$	[computer-verified] (numerically-exact Ricci, validated flat/sphere)
GTD-II KN blind to extremality (R finite on (25))	[computer-verified]
Exact closed form of den R (28)	[pending verification run]
RoleChSpec faithful: 6×3 rank-one matrix has rank 3; carrier 3-dim in all characteristics	[computer-verified] / [symbolic] (role-channel companion)
Two-state demo: R and $\mathcal{A}_c$ each constant on p_1, p_2 ; carrier separates ($\ \Delta O\ \approx 0.047$)	[computer-verified] (pseudo-arclength continuation)
Coupling coefficient = directional normal curvature N_{ab} ; output enters via $1/F_y$	[symbolic] (role-channel companion)
S -chart singularity = entropy-cover ramification = extremal boundary	¹⁴ [symbolic] (monodromy companion)
<i>DBP metric and complementarity ($n = 2$)</i>	

The GTD-II curvature (eq. (14)) was obtained by evaluating the diagonal-metric Gaussian curvature (13) on g^{II} in exact computer algebra and factoring numerator and denominator over $\mathbb{Q}[\pi, J]$; divisibility by the Davies polynomial and non-divisibility by the extremal factor were both tested exactly. The DBP identities are exact computer algebra as recorded in the companion paper.

For Kerr–Newman the GTD-II Ricci scalar was computed from a lambdified exact metric via a Ricci routine validated on flat space ($R=0$) and the unit 2–sphere ($R=2$); the Davies factor $\mathcal{D}_3$ (27) was confirmed as the unique physical vanishing factor and its $Q \rightarrow 0$ reduction to the Kerr quartic checked, and finiteness of R on the extremal locus was evaluated at representative points. The full factored den R (28) is the one quantity in this paper still pending an exact closed–form run; nothing in §5.1(i)–(ii) depends on it. The two–state demonstration (§5.3) used pseudo–arclength continuation along the simultaneous level set $\{R = \text{const}\} \cap \{\mathcal{A}_c = \text{const}\}$; the recovered pair was checked to agree in R and $\mathcal{A}_c$ to $\sim 10^{-13}$ while differing in the carrier O and in the six rank–one channels. RoleChSpec faithfulness and the directional–curvature identity N_{ab} are exact results of the role–channel companion; the boundary/ramification identification is from the monodromy companion.

References

- [1] W. Lloyd, *The Active Role-Jet Orbit Calculus for Constraint-Surface Roles* (2026). Foundational companion: active role transposition, the Active Role-Jet Orbit Theorem, channel reduction, the named output-role channels $\Lambda_P, \Lambda_D, \Lambda_S$, and the anisotropy scalar $\mathcal{A}_c$.
- [2] F. Weinhold, *Metric geometry of equilibrium thermodynamics*, J. Chem. Phys. **63**, 2479 (1975).
- [3] G. Ruppeiner, *Thermodynamics: A Riemannian geometric model*, Phys. Rev. A **20**, 1608 (1979).
- [4] H. Quevedo, *Geometrothermodynamics*, J. Math. Phys. **48**, 013506 (2007).
- [5] H. Quevedo et al., *Phase transitions in geometrothermodynamics* (2010).
- [6] G. K. Mahanta, *Thermodynamic geodesics in Bardeen regular black hole: conventional vs. modified geometrothermodynamics metrics*, Eur. Phys. J. C **85**, 1203 (2025); arXiv:2510.22398.
- [7] G. K. Mahanta, *Conventional vs. modified GTD metrics: survival of modified GTD metrics in AdS spacetime and thermodynamic ensembles*, arXiv:2603.11567 (2026).

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